

Global indicators of change in 2016

The Met Office Hadley Centre is one of a number of leading research centres around the world which monitor changes in the climate. Long-term changes in global climate indicators include temperature, snow and ice, greenhouse gases, Earth's energy budget, and sea level.

Temperature

2015 was the warmest year on record globally, surpassing the previous record set in 2014 by more than 0.1 °C. Global mean temperature at the Earth's surface in 2015 exceeded the pre-industrial average (1850-1900) by more than 1 °C for the first time.

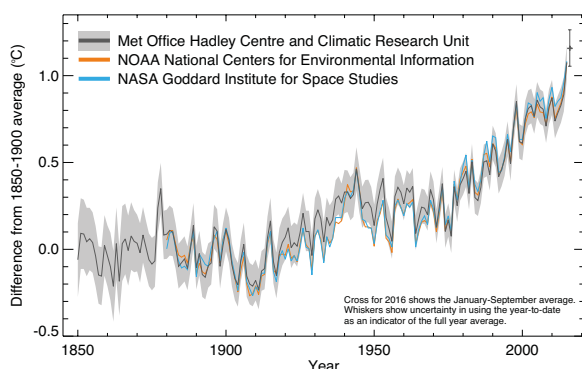


Figure 1. Average global near-surface temperature relative to 1850-1900, including uncertainties (grey shading). Source: HadCRUT4, NOAA GlobalTemp, GISTEMP (NASA).

2016 will be the third consecutive year of exceptionally high global surface temperatures. 2014 equalled the former record, 2015 broke it and 2016 is likely to be warmer still. Monthly temperatures have been at near-record or record levels since mid 2015 with the period from November 2015 to April 2016 being particularly warm.

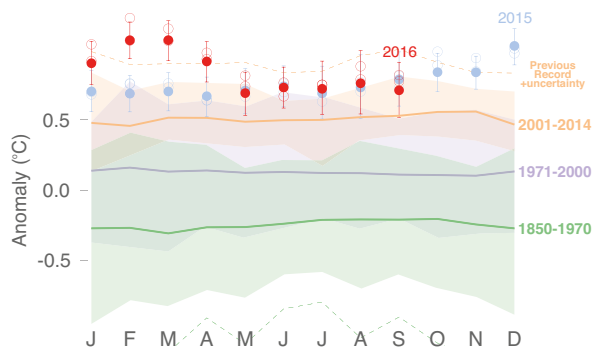


Figure 2. Monthly global temperature anomalies relative to 1961-1990 for Jan 1850-Sept 2016. Source: HadCRUT4 (filled circles), NOAA-GlobalTemp & GISTEMP (NASA) (open circles).

The global mean temperature for 2016 is on track to be within the range forecast by the Met Office in December 2015, 1.02 - 1.26 °C above the pre-industrial (1850-1900) average.

Snow and Ice

Arctic sea ice maximum extent reached a new record low in March 2016, 1.12 million km² (or 7.2%) below the 1981-2010 average. Record low monthly extents continued into June. The actual minimum extent in 2016 occurred on 10th September and was the joint 2nd lowest on record, equal to 2007 and behind 2012. The average extent for September was 3rd lowest on record, and 1.95 million km² below the 1981-2010 average.

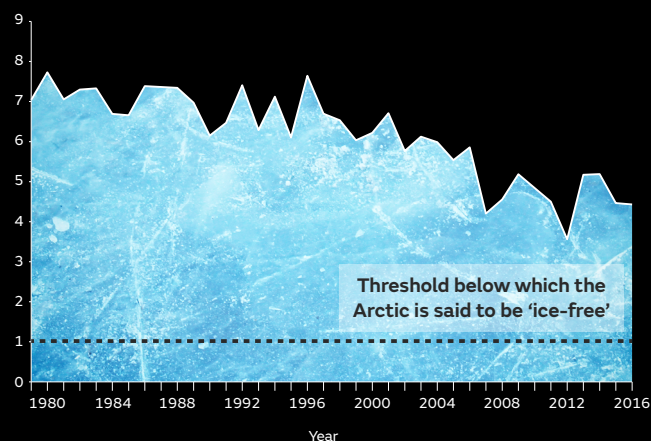


Figure 3. Mean September Arctic sea ice extent from 1979 to 2016. Source: HadISST1.2.

In the Antarctic, sea ice extent was well above the 1981-2010 average for the first half of 2015, with monthly record highs observed in January, April and May. This was followed by a sharp reversal as record low levels were recorded in August, followed by near-average levels for the rest of the year.¹

2015 was the 36th consecutive year of declining glacier mass globally, equalling 2003 with the greatest annual mass loss.¹

Snow cover in the Northern hemisphere in 2015 was below average, ranked as the 10th least extensive in the 46-year record.¹





Greenhouse Gases

Carbon dioxide, methane and nitrous oxide, the most dominant climate-forcing greenhouse gases, all reached new record high levels in 2015.¹ The atmospheric CO₂ concentration at Mauna Loa, Hawaii, increased by a record 3.05 parts per million during 2015, exceeding 400 ppm for the first time in the instrumental record, and in measurements of air trapped in ice cores for up to 800,000 years.² In May 2016, Mauna Loa recorded an average monthly value of 407.6 ppm - the highest ever monthly value.

Met Office scientists predicted that the seasonal minimum of CO₂ in the atmosphere would remain above 400 ppm for the first time in 2016, boosted by the strong El Niño.² Indeed the September minimum concentration was 400.1 ppm.

Earth's energy imbalance

Over the last four decades, the Earth has absorbed more energy from incoming solar radiation than it has emitted back to space. This energy imbalance is caused by the rising concentrations of greenhouse gases in the atmosphere.³

The oceans have absorbed more than 90% of this excess energy. Whilst surface warming is most pronounced, warming has been observed throughout the ocean depth. The remainder of the energy imbalance is accounted for in warming of the atmosphere and continents, and in melting ice.⁴

The oceans store so much heat because water has a high heat capacity, and heat absorbed at the surface is transported to greater depths through ocean circulation.

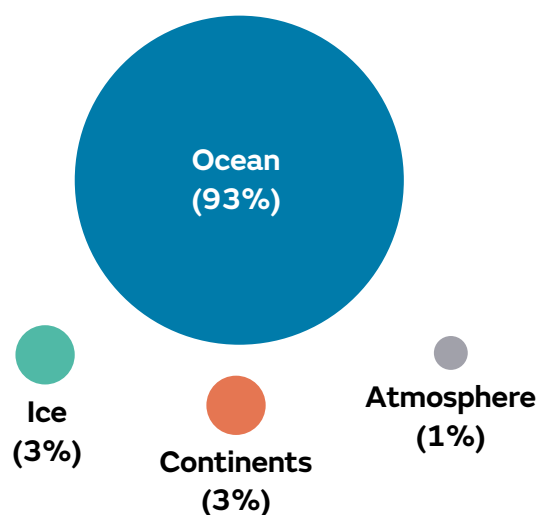
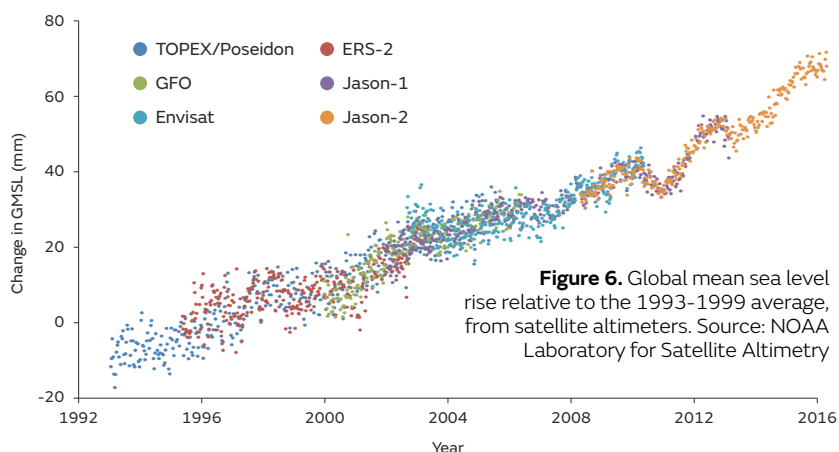


Figure 5. Excess energy absorbed by each component of the climate system since the 1970s.³

Rising sea levels

As the climate warms, land ice melt and the thermal expansion of water causes global sea level rise. Global mean sea level (GMSL) has risen by around 1.7 mm per year over the last century, increasing to 3.3 mm per year since the early 1990s. In 2015, GMSL averaged over the year was the highest in the satellite record and 7cm above the 1993-1999 mean.¹

Even with stringent emissions reductions, GMSL will continue to rise throughout the 21st Century and beyond, due to increased ocean warming and land ice mass loss.³



References: ¹Blunden & Arndt, 2016; ²Betts et al., 2016; ³IPCC AR5, 2013; ⁴Johnson et al., 2016.
References in bold indicate publications with Met Office authors/editors.